Does crossing an ISA change the relative cost of database operators?

No. And that refutes the stated motivation for per-operator capability modelling.

Measured with `scripts/collection/operator_probe.py` (reps=9, both records passing the repeatability gate) on:

	dev	prod
machine	Apple M1	AMD EPYC 9V74
ISA	**arm64**	**x86_64**
cores	8	4
PostgreSQL	15.17 (Homebrew)	15.19 (Debian, `postgres:15`)
provenance	laptop	GitHub Actions runner, `.github/workflows/x86-operator-probe.yml`

The measurement

axis	M1 (arm64) tuples/s	EPYC (x86_64) tuples/s	ratio
seqscan	11,487,849	8,753,444	1.312
hashjoin	4,187,714	3,784,817	1.106
sort	4,610,457	4,000,748	1.152
hashagg	3,458,930	2,317,419	1.493
nestloop	842,971	770,706	1.094
parallel_speedup	2.625	2.211	1.187

Cross-ISA operator-ratio spread: max/min = 1.365×.

Why this is a negative result

A per-machine **scalar** model multiplies dev runtime by one number. It is wrong exactly to the extent that the true ratio differs *between operators* — if every axis scaled by the same factor, a scalar would be sufficient and per-operator modelling would be pointless.

The argument for per-operator rates, as stated earlier in this project, was that an ISA boundary changes the *relative* cost of operators, so cross-ISA needs per-query capability information. **That argument does not survive measurement:**

	operator-ratio spread
arm64 → x86_64 (different ISA, different vendor, different core count)	**1.365×**
worst pair *within* the arm64 container grid (`c8m1g` → `c1m3g`, same ISA, same host)	**1.371×**

Crossing the ISA boundary perturbs the operator mix no more than changing a CPU quota on one machine does. So per-operator weighting cannot be expected to rescue cross-ISA prediction, and the 59.7% irreducible floor measured on laptop → EC2 (laptop_to_ec2.md) is **not** caused by operator-mix divergence.

This is consistent with the grid ablation, which already found per-operator rate weighting worth only −0.08 pp (CI [−0.73, +0.44]) against a single scalar carrying the same parallelism information. Two independent experiments now say the same thing: **per-operator *rates* are not the useful part of this model. The parallelism term and its plan-structure localisation are.**

What must therefore explain the cross-ISA floor

Not the ISA, and not operator mix. The remaining candidates, in order of evidence:

1. **Storage and buffer-pool regime.** The EC2 instances were EBS-backed (network storage); the laptop is local NVMe. Over-prediction in laptop → container already correlated with per-query I/O fraction (Spearman r = +0.302), and the probe has no axis that measures the storage path — the `io_seqscan` axis built for it tracks CPU quota (+0.762) over memory (+0.548) and so does not isolate storage.
2. **Per-query variance that no per-machine quantity can express.** On EC2 the scaling factor spans 18–43× *within a single machine* across queries. Per-operator rates are still a per-*machine* quantity; they reweight by plan composition but cannot represent per-query effects of that magnitude.
3. **Engine build and configuration differences** (Homebrew vs Debian builds, and an inferred rather than measured `max_parallel_workers_per_gather` on EC2).

Scope

- **One x86 machine, one arm64 machine.** Two machines cannot establish a distribution; this refutes a strong claim ("ISA changes operator mix substantially") rather than establishing its converse with confidence.
- A CI runner is shared hardware. The record passed the repeatability gate (bootstrap SE of the median, all axes ≤ 0.08) but its neighbours are not controlled.
- The comparison is signature-level. It says the *mechanism* proposed for beating the floor is absent; the runtime-prediction test on the same pair is reported separately once the workload collection completes.